

[Knowledge Representation Issues, Mapping, Frame Problem]

Predicate Logic

Predicate logic is an extension of Propositional logic that allows us to represent → Objects Properties of objects
Relationship b/w objects Quantifiers

• Propositional Logic

It is the simplest form of logic used in AI. It deals with statements (propositions) that are either TRUE or FALSE, but not both.

e.g → Example of Proposition

- Ram is a student ✓
- Close the door X
- $2+2 = 4$ ✓

• Atomic and Compound Propositions

a) Atomic Proposition → A single statement

[P: It is raining] Cannot be broken further

b) Compound → Formed by combining atomic propositions using logical operators.

• Logical connectives $\vee$ → OR → disjunction

$\neg, \sim$ → not $(\rightarrow)$ → implies → implication

$\wedge$ → and $(\leftrightarrow)$ → iff → biconditional

Conjunction

Truth Table

i) $\neg$

P	$\neg P$
T	F
F	T

ii) AND ($\wedge$)

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

iii) OR ($\vee$)

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication ($\rightarrow$)

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Only false when P is true and Q is false.

v) Biconditional ($\leftrightarrow$)

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

When both same true
different false

Well formed formula

when it follows syntax rules.

$$(P \wedge Q) \rightarrow R \checkmark$$

$$P \wedge \rightarrow Q \times$$

Tautology, Contradiction, Contingency

always true	always false	Sometimes true
$P \vee \neg P$	$P \wedge \neg P$	Sometimes false
		$P \wedge Q$

Logical equivalence

↪ two formulas are equivalent if they have same truth table results.

$$[P \rightarrow Q \equiv \sim P \vee Q]$$

e.g. → If it rains, the road is wet

P = It rains

$P \rightarrow Q$

Q = Road is wet

It is raining and cold

P = raining

Q = cold

$P \wedge Q$

It is not hot

$\sim P$

Q Check whether the following formula is a tautology

$$((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$$

If P implies Q and
 Q implies R then

P implies R

this is always true

Predicate logic $\rightarrow$ Quantifiers

i) Universal Quantifier ($\forall$)

means "for all"

all students are intelligent

$$\forall x [\text{Student}(x) \rightarrow \text{Intelligent}(x)]$$

ii) Existential Quantifier ($\exists$)

means "there exists / some"

there exists a student who is intelligent

$$\exists x [\text{Student}(x) \wedge \text{Intelligent}(x)]$$
